

Xuming Huang

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Systems for AI; Edge Inference; Operating Systems; AI Compilers

Education

University of Wisconsin–Madison

B.S. in Computer Sciences

GPA: **3.96**/4.00

Madison, WI

Jan. 2025 – May 2027 (Expected)

Stanford University

Summer Session, Visiting Undergraduate

GPA: **4.00**/4.00

Stanford, CA

Jun. 2025 – Aug. 2025

University of Shanghai for Science and Technology

Undergraduate study in Computer Science and Technology

GPA: **3.82**/4.50

Shanghai, China

Sept. 2021 – Jul. 2024

Research Experience

Edge AI Systems Optimization, WukLab, UC San Diego

Jan. 2026 – Present

Undergraduate Researcher advised by Prof. Yiyang Zhang

- Developed a JIT (just-in-time) NPU runtime system that parallelizes progressive graph compilation on CPU with LLM inference on NPU, and organizes copy-and-grow KV cache as context expands; reduced foreground compilation time by 91.98%, saved 33.05% averaged KV memory, and preserved model quality and decode
- Optimized the OpenVINO NPU compiler by caching sorted dependency and consumer vectors; reduced CPU sorting time by 94.2%, and contributed [PR #348](#) to the Intel NPU Compiler repository
- Developed a CPU/GPU/NPU benchmarking methodology to characterize operator and inference-stage affinity across edge devices; identified hardware-dependent placement preferences to guide heterogeneous scheduling

Operating Systems for AI Inference, UW Madison [\[Page\]](#)

Jan. 2026 – May 2026

Honors Course Research Project advised by Prof. Michael Swift

- Evaluated filesystem interference on LLM inference using controlled background file reads and fixed inference workloads; measured 17.1% higher median latency than baseline
- Investigated inference slowdown using Windows PerfMon; observed stable working sets and page-fault counts plateauing near 29.5K despite rising latency, supporting shared-resource contention over working-set thrashing

LinuxGuard: AI for System Security, UW–Madison [\[Page\]](#)

Jan. 2025 – Nov. 2025

Undergraduate Researcher advised by Prof. Remzi Arpaci-Dusseau and Dr. Vinay Banakar

- Built an LLM-driven pipeline over 50K+ Linux kernel commits to extract bug patterns and synthesize Clang-Tidy analyzers; reduced manual vulnerability triage from weeks to hours
- Developed a multi-stage checker synthesis pipeline with K-means clustering over 10K+ code patterns and compiler-feedback repair; increased valid checkers 4x while achieving 73% precision
- Engineered a full-kernel scanning and validation harness to evaluate checker failures and false positives; uncovered 43 long-latent kernel bugs with an average age of 4.7 years

ML-Guided Scheduling for Heterogeneous Compute Systems, USST [\[Paper\]](#)

Jan. 2024 – Sept. 2026

Coauthor | Undergraduate Researcher advised by Prof. Xing Hu

- Evaluated an ML-guided scheduler that predicts workload–hardware affinity using calibrated DCN/XGBoost classifiers and places tasks across heterogeneous CPU/GPU pools based on runtime, energy, and resource availability
- Contributed to experimental validation and performance analysis on MIT Supercloud traces, achieving 97.1% resource-matching accuracy, 46.5% lower response time, and 36.8% lower modeled energy than Meta-RHDC in 5,000-task simulations

Publications

[3] CASH: Confidence-Calibrated Deep Feature Crossing for Classification-Aware Heterogeneous Task Scheduling.

Chuanlin Jian, Yuanchen Sun, Xiangcheng Liu, Xuming Huang, Samaneh Beheshti Kashi, and Xing Hu
Computers, Materials & Continua, 89(2), Article 29, 2026. [DOI](#)

[2] A Method for Laryngoscopic Image Classification: Leveraging Self-supervised BEiT-style Pretraining.

Jie Kong, Haotian Wu, Xuming Huang, Liangjie Wu, and Xing Hu

In *7th International Conference on Frontier Technologies of Information and Computer (ICFTIC)*, pp. 701–706, 2025. [DOI](#)

[1] Near-infrared tuneable filter based on polymer dispersed liquid crystals.

Ruimei Zeng, Rongbo Bi, Junyang Zhu, Xuming Huang, and Qi Wang

Micro & Nano Letters, 20(1), e70007, 2025. [DOI](#)

Teaching Experience

ECE 252, Department of Electrical and Computer Engineering, UW–Madison
ICPC (In-Class Peer Coach)

Sept. 2026 - Dec. 2026

- Support discussion session and hold office hours for ECE 252

Selected Projects

Algorithms and AI Systems Learning Resources [\[Algo, AI Sys\]](#)

Jan. 2024 – Present

Open-source Author and Maintainer

- Developed and maintained 20+ algorithm and data-structure implementations across 6+ topics, with visualizations and practice templates; reached 100+ GitHub stars
- Implemented Transformer and GPT-2 from scratch in PyTorch, covering multi-head attention, encoder-decoder architecture, optimization, and training

Heap Allocator

Jun. 2025 – Aug. 2025

Systems Project during Stanford Summer Session

- Implemented free-block management, coalescing, and boundary tags; explored fragmentation and allocator behavior through randomized stress tests and memory-debugging tools

Work Experience

Cool AI Technology

Jul. 2024 – Sept. 2024

Technical R&D and Product Development Intern, Shanghai

- Redesigned the company’s web interface using Next.js and Tailwind CSS, improving page load time by 45% and increasing user engagement metrics by 30% across 5000+ daily active users
- Architected a scalable FastAPI backend for LLM integration, handling 1,000+ concurrent requests and reducing API response time from 3s to 800ms
- Deployed Prompted Agents for AI-Hub platform, serving 20+ enterprise clients, generating \$10K in new revenue within the first month of launch

Technical Skills

Languages: Python, C/C++, Java, JavaScript, Verilog, assembly, Swift

Systems and tools: Linux, OpenVINO, LLVM/Clang, Git, Docker, GDB, Valgrind

Methods: Performance measurement, compiler/runtime experiments, model fine-tuning, correctness evaluation

Honors and Awards

- Dean’s List, University of Wisconsin–Madison
- Presidential Scholarship, University of Shanghai for Science and Technology (Top 1%)